

DSP Hardware Overview

2.1.5 DSP hardware and processors

2.1.5 DSP Hardware and Processors (Detailed Explanation)

English + Nepali Mixup (No Romanize)

Introduction

DSP (Digital Signal Processing) Hardware भनेको digital signals लाई fast, accurate, र real-time मा process गर्न बनाइएको specialized electronic hardware हो।

साधारण CPU ले पनि signal processing गर्न सक्छ, तर complex mathematical operations जस्तै:

- Filtering
- FFT (Fast Fourier Transform)
- Compression
- Image Processing
- Audio Processing
- Video Processing

जस्ता tasks धेरै fast गर्न **DSP Processor** प्रयोग गरिन्छ।

Simple शब्दमा,

DSP Hardware = Specialized hardware that performs digital signal processing efficiently.

What is a DSP Processor?

DSP Processor भनेको एउटा **special-purpose microprocessor** हो जुन signal processing applications का लागि design गरिएको हुन्छ।

General CPU ले documents, games, browsers आदि चलाउँछ।

DSP Processor ले भने:

- Audio
- Voice
- Image
- Video
- Radar

- Medical Signals

जस्ता signals process गर्छ।

Why Normal CPU is Not Enough?

Example:

Imagine एउटा mobile phone मा call चलिरहेको छ।

Normal CPU ले

- Operating system
- Camera
- WiFi
- Apps

सबै handle गरिरहेको हुन्छ।

यदि voice processing पनि CPU ले मात्र गर्यो भने:

- Delay आउन सक्छ
- Noise remove गर्न slow हुन्छ
- Echo cancellation ढिलो हुन्छ

त्यसैले DSP processor use गरिन्छ।

DSP Hardware Components

DSP Hardware मुख्य रूपमा निम्न components बाट बनेको हुन्छ।

1. Input Device

Signal collect गर्ने device।

Examples

- Microphone
- Camera
- Temperature Sensor
- ECG Sensor
- Pressure Sensor

Example

Microphone बोलाइलाई analog signal मा convert गर्छ।

2. ADC (Analog to Digital Converter)

Real-world signal analog हुन्छ।

Computer ले analog बुझ्दैन।

ADC ले

Analog Signal

↓

Digital Signal

मा convert गर्छ।

Example

Voice → ADC → Digital Samples

3. DSP Processor

यो system को brain हो।

यसले mathematical processing गर्छ।

Examples

- Noise Removal
 - Filtering
 - Compression
 - FFT
 - Feature Extraction
 - Voice Recognition
-

4. Memory

DSP Processor लाई data store गर्न memory चाहिन्छ।

Memory Types

RAM

Temporary storage

Store गर्छ

- Input samples
- Intermediate calculations
- Output samples

ROM / Flash

Permanent storage

Store गर्छ

- DSP Program
- Algorithms
- Firmware

5. DAC (Digital to Analog Converter)

Processing पछि signal digital हुन्छ।

Speaker ले analog मात्र बजाउन सक्छ।

त्यसैले

Digital

↓

Analog

conversion DAC ले गर्छ।

Example

Digital Audio

↓

DAC

↓

Speaker

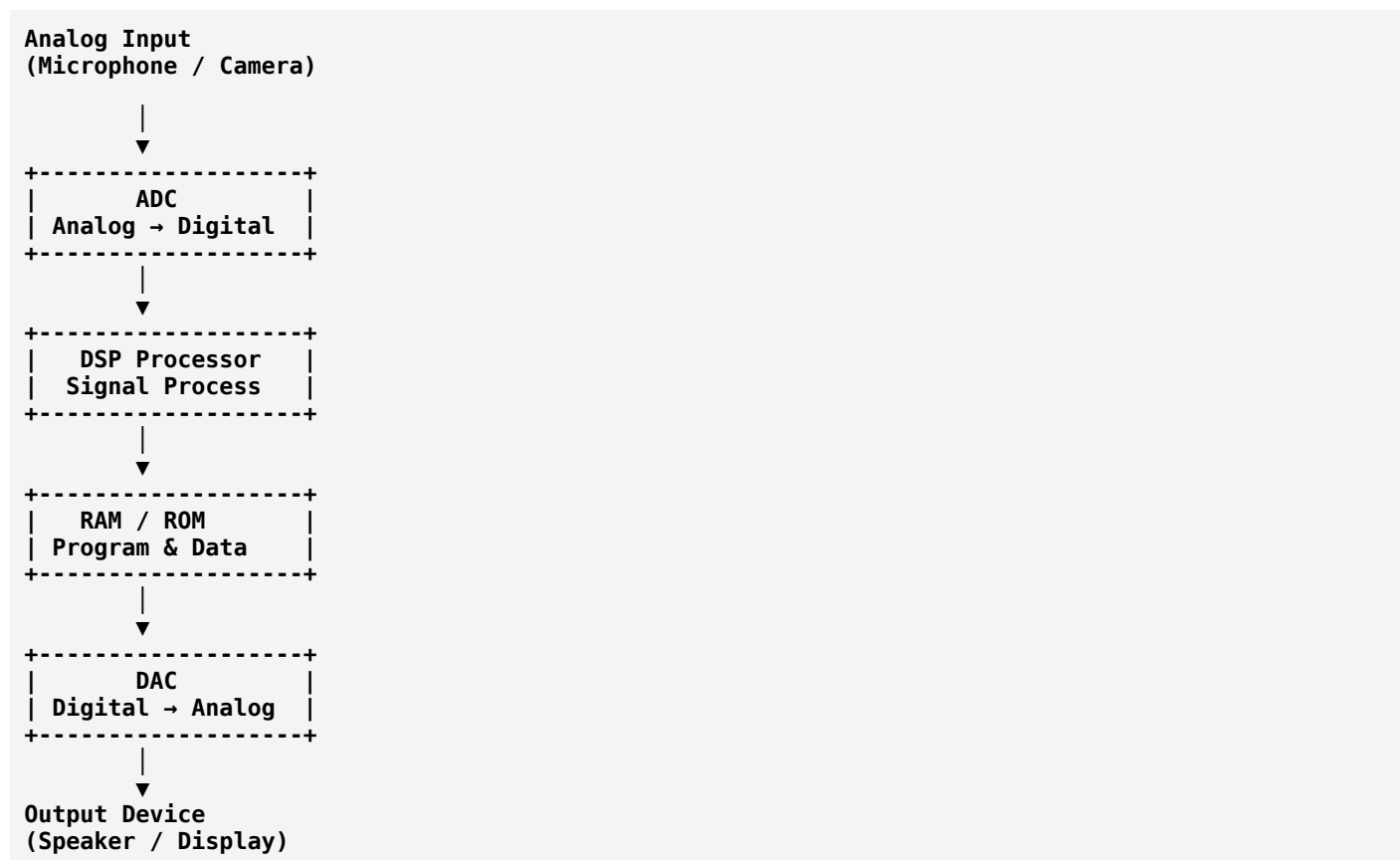
6. Output Device

Final processed signal देखाउने वा बजाउने device।

Examples

- Speaker
- Display
- Monitor
- Printer
- Medical Display

Overall DSP Hardware Block Diagram



Architecture of DSP Processor

DSP Processor भित्र पनि धेरै important units हुन्छन्।

1. ALU (Arithmetic Logic Unit)

काम

- Addition
- Subtraction
- Comparison
- Logical operations

2. Multiplier Unit

DSP मा multiplication धेरै हुन्छ।

Example

Digital Filter

$$y = ax + by + cz$$

यसका लागि fast multiplier चाहिन्छ।

DSP Processor मा dedicated multiplier हुन्छ।

3. MAC Unit (Multiply-Accumulate Unit)

DSP को सबैभन्दा important feature।

Operation

$$(A \times B) + C$$

एकै clock cycle मा हुन्छ।

Example

$$2 \times 5 + 8$$

=

$$18$$

Normal CPU

- Multiply
- Store
- Add

तीन चरण लाग्न सक्छ।

DSP

एकै operation मा गर्छ।

त्यसैले धेरै fast हुन्छ।

4. Program Memory

Instruction store गर्छ।

Example

- FFT Program
 - Audio Filter
 - Image Enhancement
-

5. Data Memory

Signal samples store गर्छ।

6. Address Generator

Memory location automatically manage गर्छ।

7. Input Output Controller

External devices सँग communicate गर्छ।

Examples

- ADC
 - DAC
 - USB
 - UART
 - SPI
 - I2C
-

Features of DSP Processor

1. High-Speed Processing

Millions of calculations प्रति second गर्न सक्छ।

2. Parallel Processing

एकै समयमा धेरै operations गर्छ।

3. Real-Time Processing

Input आउँदा तुरुन्त process गर्छ।

Example

Live Audio

4. Low Power Consumption

Battery devices मा useful हुन्छ।

Example

- Mobile Phone
 - Hearing Aid
-

5. Dedicated Hardware

Signal processing का लागि optimized हुन्छ।

6. High Accuracy

Floating-point वा Fixed-point calculations support गर्छ।

Types of DSP Processors

1. Fixed-Point DSP

Integer numbers प्रयोग गर्छ।

Advantages

- Cheap
- Fast
- Low Power

Applications

- Washing Machine
 - Calculator
 - TV
 - MP3 Player
-

2. Floating-Point DSP

Decimal calculations support गर्छ।

Advantages

- High Precision
- Scientific calculations

Applications

- Radar
- Medical Equipment
- Satellite
- Scientific Instruments

Popular DSP Processor Manufacturers

Company	DSP Processor
Texas Instruments	TMS320 Series
Analog Devices	SHARC, Blackfin
Qualcomm	Hexagon DSP
NXP	StarCore DSP
Microchip	dsPIC Series

Applications of DSP Hardware

Audio Processing

- Noise Removal
- Echo Cancellation
- Music Enhancement

Image Processing

- Face Detection
- Camera Enhancement
- Image Compression

Video Processing

- Video Compression
- Video Stabilization
- Streaming

Medical Equipment

- ECG Analysis
- EEG Analysis
- Ultrasound
- MRI

Communication Systems

- Mobile Networks
- WiFi
- Bluetooth
- Satellite Communication

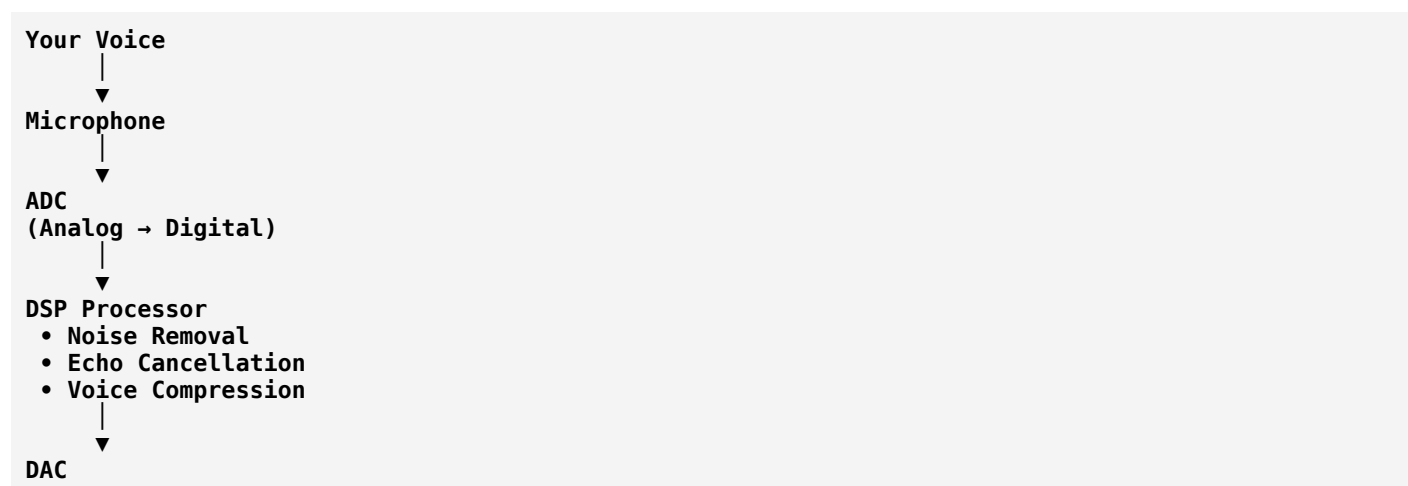
Automotive Systems

- Voice Control
- ADAS
- Radar
- Parking Sensors

Industrial Automation

- Motor Control
- Robotics
- Process Monitoring

Real-Life Example: Smartphone Voice Call



(Digital → Analog)



Speaker

Explanation:

- तपाईं बोल्दा **microphone** ले आवाजलाई analog signal मा capture गर्छ।
- **ADC** ले त्यसलाई digital signal मा convert गर्छ।
- **DSP processor** ले background noise हटाउँछ, echo reduce गर्छ, र voice लाई compress गर्छ।
- **DAC** ले processed digital signal लाई फेरि analog मा बदल्छ।
- अन्त्यमा **speaker** बाट स्पष्ट आवाज सुनिन्छ।

Advantages of DSP Hardware

- Very fast processing
- Real-time operation
- High accuracy
- Low power consumption
- Reliable performance
- Efficient mathematical computation
- Excellent for multimedia applications

Disadvantages of DSP Hardware

- General CPU भन्दा महँगो हुन सक्छ।
- Programming तुलनात्मक रूपमा complex हुन्छ।
- Specialized knowledge आवश्यक पर्छ।
- Simple applications का लागि overkill हुन सक्छ।

Interview / Exam Points (Very Important)

Definition: DSP Hardware भनेको digital signals लाई fast र efficient रूपमा process गर्न design गरिएको specialized hardware हो।

Main Components:

- Input Device
- ADC
- DSP Processor
- RAM / ROM

- DAC
- Output Device

Important Features:

- High-Speed Processing
- MAC Unit
- Parallel Processing
- Real-Time Processing
- Low Power Consumption
- High Accuracy

Applications:

- Audio Processing
- Image Processing
- Video Processing
- Communication Systems
- Medical Equipment
- Industrial Automation
- Automotive Systems

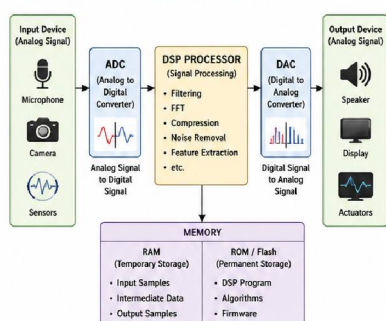
Remember: DSP hardware को मुख्य उद्देश्य **real-time** मा **digital signals** लाई छिटो, **accurate**, र **efficient** तरिकाले **process** गर्नु हो। Specialized DSP processors मा रहेको **MAC (Multiply-Accumulate) Unit**, dedicated memory architecture, र parallel processing capability ले यसलाई सामान्य CPU भन्दा signal processing का काममा धेरै प्रभावकारी बनाउँछ।

image illustration for pdf size

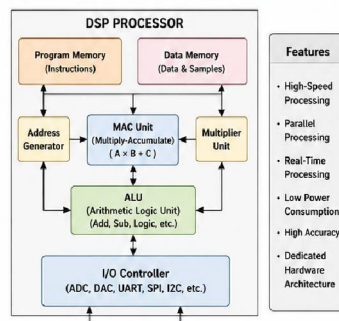
2.1.5 DSP HARDWARE AND PROCESSORS

DSP (Digital Signal Processing) hardware is a specialized hardware system designed to process digital signals efficiently and in real-time.

1. BASIC DSP HARDWARE BLOCK DIAGRAM



2. INTERNAL ARCHITECTURE OF DSP PROCESSOR



3. SIGNAL FLOW IN DSP SYSTEM (EXAMPLE: VOICE PROCESSING)



4. TYPES OF DSP PROCESSORS

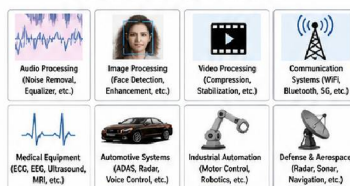
(A) Fixed-Point DSP	(B) Floating-Point DSP
- Uses integer numbers - Fast, low power, low cost - Suitable for simple to moderate precision tasks Applications: - Consumer Electronics - Washing Machine - MP3 Player - TV Set-top Box	- Uses decimal (floating) numbers - High precision - Suitable for complex calculations Applications: - Radar Systems - Medical Equipment - Satellite Systems - Scientific Instruments

5. POPULAR DSP PROCESSOR FAMILIES

Company	DSP Processor Series
Texas Instruments	TMS320 Series
Analog Devices	SHARC, Blackfin Series
Qualcomm	Hexagon DSP
NXP	StarCore DSP
Microchip	dsPIC Series



6. APPLICATIONS OF DSP HARDWARE



7. ADVANTAGES AND DISADVANTAGES

Advantages	Disadvantages
✓ Very fast processing ✓ Real-time operation ✓ High accuracy ✓ Low power consumption ✓ Reliable performance ✓ Efficient mathematical computation ✓ Excellent for multimedia applications	✗ Generally more expensive than normal CPU ✗ Programming is complex ✗ Specialized knowledge is required ✗ Not suitable for simple tasks (overkill)

Summary: DSP hardware uses ADC to convert analog signals to digital, processes them using a powerful DSP processor with specialized units like MAC, ALU, and Multiplier, stores data in memory, converts the result back to analog using DAC, and delivers the output through output devices. It is widely used in real-time signal processing applications.